↓ ↓

[~~1~~, ~~2~~, ~~3~~, 5, 6, 7, 8, 9, 10, 1, 2, 3]

↓ target = 3

0 1 2 3 4 5 6 7 8

[5, 6, 7, 8, 9, 10, 1, 2, 3]

low = 0 high = 8

mid = $\frac{8}{2}$ = 4 Disc. the LH.

low = 5 high = 8

mid = $\frac{13}{2}$ = 6 Disc. the LH.

low = 7 high = 8

mid = $\frac{15}{2}$ = 7 Disc. the LH.

low = 8 high = 8

mid = 8

```
if arr[mid] >= arr[low]: LH is sorted.
    if tar >= arr[low] && tar <= arr[mid]:
        search in the LH.
    else:
        search in the RH.
else:    RH is sorted.
    if tar >= arr[mid] && tar <= arr[high]:
        search in the RH.
    else:
        search in the LH.
```

arr[low] .... arr[mid] .... arr[high]

TC: O(log n)

AS: O(1).

N = 4 M = 2

[10, 20, 10, 30].

[10]. [20, 10, 30] → 10, $\boxed{60}$

[10, 20] [10, 30] → 30, $\boxed{40}$

[10, 20, 10] [30] → $\boxed{40,}$ 30

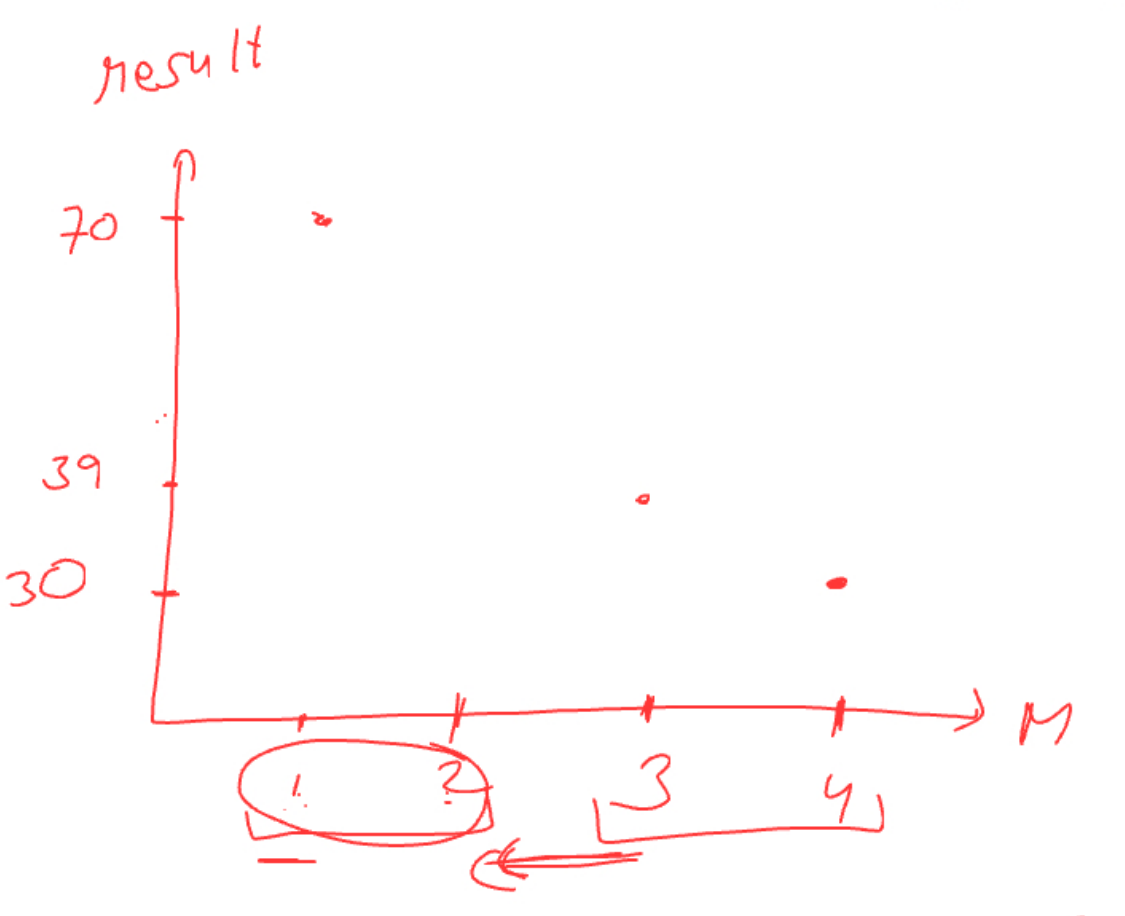
O/P = 40

$^{n}C_{m-1} \cdot n$

low = 30 high = 70

M = N M = 1

res = ~~50~~ ~~44~~ ~~41~~ 40



ns = ~~0~~ 1

nl = ~~0~~ ~~10~~ ~~30~~ 10

[10, 20, 10, 30]
1 1 2 3

mid = $\frac{100}{2}$ = 50

[10, 20, 10, 30]
1 1 1 2

Disc. the RH

---

low = 30 high = 49

mid = $\frac{79}{2}$ = 39

[10, 20, 10, 30]
1 1 1 2

Disc. the LH

---

low = 40 high = 49

mid = $\frac{89}{2}$ = 44

[10, 20, 10, 30]
1 1 1 2

Disc. the RH

---

low = 40 high = 43

mid = $\frac{83}{2}$ = 41

Disc. the RH.

---

low = 40 high = 40.

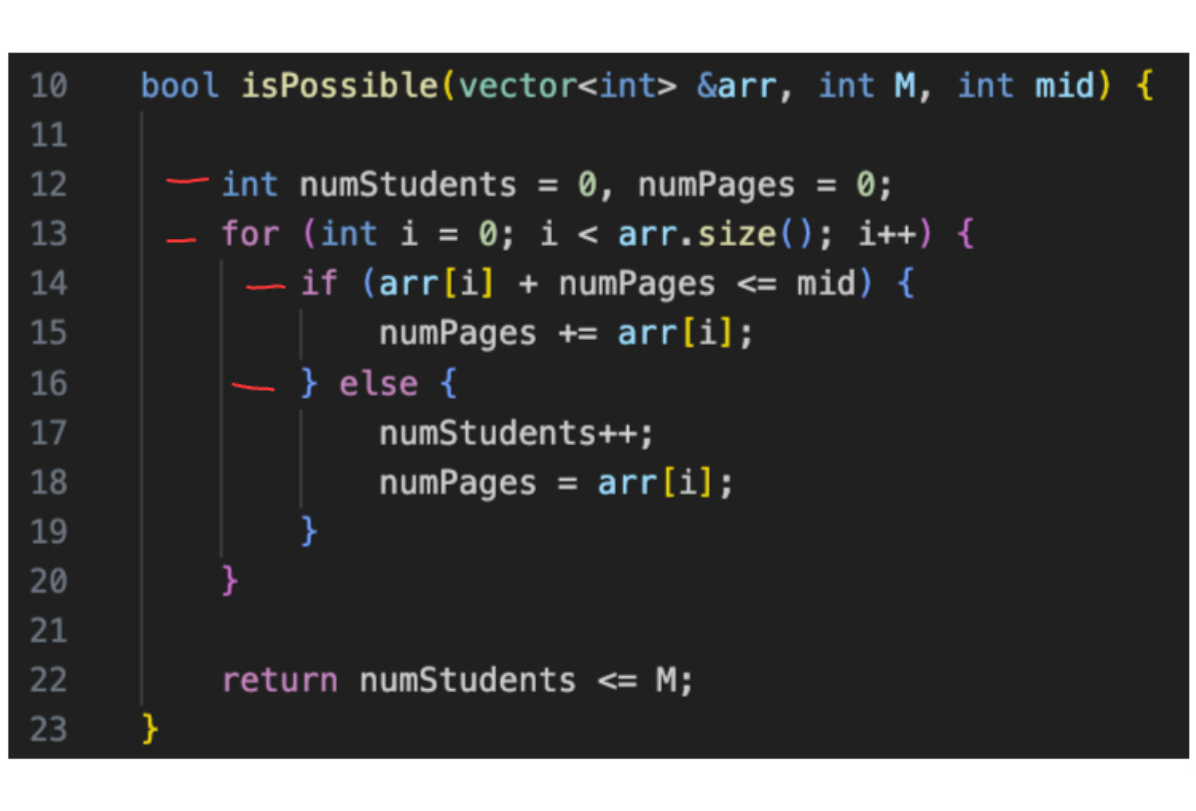
mid = 40

Disc. the RH.

---

low = 40 high = 39.

STOP!



```
bool isPossible(vector<int> &arr, int M, int mid) {

    int numStudents = 0, numPages = 0;
    for (int i = 0; i < arr.size(); i++) {
        if (arr[i] + numPages <= mid) {
            numPages += arr[i];
        } else {
            numStudents++;
            numPages = arr[i];
        }
    }

    return numStudents <= M;
}
```

[10, 20, 10, 30]

M = 2

mid = 39.

ns = ~~0~~ ~~1~~ 2

nl = ~~10~~ ~~30~~ ~~30~~ ~~10~~ 30

ns = 0 nl = 10

TC = O(n * log(sum))